

askADZA Full Commercialization Cost Summary

v7 — Commercial pricing ladder + allowance-based unit economics; R1 CoT only on Agribusinesses and Integrated; Tavily Project + capped BQ + local rerank

Capacity traffic: 5,000 API req/day (150k/mo · 450k/qtr) for shared infra sizing

Profitability lens: Per-plan COGS at query allowance (not 5 req/user/day)

R1 CoT: Agribusinesses & Financial Institutions + Integrated Account only (+~\$0.01/req)

Web (RAG): Tavily Project (~\$30/mo, ~4k credits). Free = no web. Mining on a separate key/budget.

Other stack: Local BGE reranker on Railway; Qdrant HNSW m=8, 1580-d, PQ (~\$400/mo); BQ Free=off; paid BQ capped 250 MiB/job bronze → 100 MiB marts.

1. Executive headlines

Lower tiers (Farmers → NGOs) show strong contribution margins at full allowance. Agribusinesses and Integrated are tighter because DeepSeek R1 CoT applies only on those two plans. Platform COGS at capacity stays near Scenario A + Tavily Project unless Agri/Integrated traffic share is large.

Lens	Figure	Notes
Platform A + Project (mostly non-R1)	~\$1.28k/mo · ~\$3.8k/qtr	5k req/day capacity
Platform if ~20% Agri/Integrated + R1	~\$1.6k/mo	+\$0.002/req blended R1 uplift
Full R1 on 100% traffic (not planned)	~\$2.83k/mo	Avoid — R1 is last two plans only
Tightest paid margin (Integrated @ 2k soft)	~26% gross	~\$12.79 contribution / \$49.99
Strongest paid margin (Farmers @ 50)	~86–88% gross	~\$2.56–2.64 contribution / \$2.99

2. Commercial pricing ladder

Plan	Price	Target user	Queries / mo
Free	\$0	Curious users, casual explorers, prospective subscribers	5
Farmers, Cooperatives & Communities	\$2.99	Smallholder farmers, cooperative members, community organisers	50
Government & Public Institutions	\$9.99	Policy analysts, planning officers, ministry staff	200
Foundations, NGOs & Development Partners	\$14.99	Program officers, M&E leads, grant managers	400
Agribusinesses & Financial Institutions	\$24.99	Risk analysts, sourcing managers, agri-finance officers, investors	800
Integrated Account	\$49.99	Consultants, researchers, multi-mandate users	Unlimited* (~2,000 soft)

* Integrated Account queries are subject to fair use; soft cap ~2,000/month for COGS planning.

3. Feature / model matrix by plan

Plan	Generate model	BQ	Tavily	R1 CoT
Free	Llama 3.1 8B	No	No	No
Farmers, Cooperatives & Communities	Llama 3.1 70B	Yes capped	Yes	No
Government & Public Institutions	Qwen2.5-72B	Yes capped	Yes	No
Foundations, NGOs & Development Partners	Qwen2.5-72B	Yes capped	Yes	No
Agribusinesses & Financial Institutions	Qwen3-235B Instruct	Yes capped	Yes	Yes
Integrated Account	Qwen3-235B Instruct	Yes capped	Yes	Yes

Pipeline (decompose / NL2SQL): Llama 3.1 8B for all plans. Rerank: local BAAI/bge-reranker-base on Railway (\$0 API).

4. Per-request COGS bands

Bronze BQ caps + Tavily Project allocated across 150k req/mo capacity. Farmers/Gov/NGO use cheaper generate models than Integrated-path A; A (~\$0.0086) is the planning ceiling for non-R1 paid.

Band	\$/request	Applies to
Free	~\$0.005	Free only (no BQ, no Tavily)
Paid non-R1 (ceiling)	~\$0.007–0.0086	Farmers, Government, NGOs
Paid + R1 (Agri / Integrated)	~\$0.0186	Last two plans only

4.1 Scenario A core itemization (before R1)

Line	\$/request	\$/month @ 150k
LLM pipeline (8B)	~\$0.00055	~\$83
LLM generate (Qwen3-235B Instruct ceiling)	~\$0.0012	~\$180
Local cross-encoder rerank	\$0	\$0
BigQuery (capped bronze)	~\$0.0020	~\$300
Railway API+Redis+rerank	~\$0.00053	~\$80
Qdrant m=8 / 1580d / PQ	~\$0.00267	~\$400
ColPali pilot	~\$0.00067	~\$100
Langfuse self-host	~\$0.0005	~\$75
GCS / egress	~\$0.0002	~\$30
Core A subtotal	~\$0.0084	~\$1.25k
Tavily Project allocated	~\$0.0002	~\$30
A + Project	~\$0.0086	~\$1.28k
R1 CoT (Agri/Integrated only)	+~\$0.01	Only on that traffic share

5. Unit economics at full allowance (worst case)

Primary profitability lens: COGS if the subscriber uses their full monthly allowance. Contribution = price – COGS. Free is an acquisition cost, not a margin target.

Plan	Price	Max queries	Est. COGS @ allowance	Contribution	Gross margin
Free	\$0	5	~\$0.03	~\$0.03	n/a (acquisition)
Farmers	\$2.99	50	~\$0.35–0.43	~\$2.56–2.64	~86–88%
Government	\$9.99	200	~\$1.40–1.72	~\$8.27–8.59	~83–86%
NGOs	\$14.99	400	~\$2.80–3.44	~\$11.55–12.19	~77–81%
Agribusinesses	\$24.99	800	~\$14.88	~\$10.11	~40%
Integrated	\$49.99	2,000 soft	~\$37.20	~\$12.79	~26%

- Shared infra is embedded in per-req rates at 150k/mo scale; under-utilization raises fixed \$/req until traffic catches up.
- Agri/Integrated margins are tightest because of R1; without R1 those tiers would resemble NGO-class margins at volume.
- Enforce Integrated soft cap (~2,000/mo) in product — uncapped R1 usage can erase the ~\$12.79 contribution.

6. Platform rollup (capacity)

Scenario	\$/mo @ 5k req/day	\$/qtr	When
A + Project (mostly non-R1)	~\$1.28k	~\$3.8k	Default capacity budget
A-marts + Project	~\$1.09k	~\$3.3k	After NL2SQL faces marts
~20% Agri/Integrated + R1	~\$1.6k	~\$4.8k	Illustrative paid mix
B = 100% R1 (not planned)	~\$2.83k	~\$8.5k	Do not use as primary budget
Streamlit ops	~\$25	~\$75	Dev/QA only; not in 5k API

R1 is not applied to 100% of traffic. Example: if 20% of requests are Agri/Integrated with R1, blended uplift $\approx 0.2 \times \$0.01 = +\$0.002/\text{req}$ → platform nearer ~\$1.6k/mo than full Scenario B.

7. Tavily Project (RAG web)

Plan	Tavily / web fallback
Free	No
All paid plans	Yes (eligible)

Hit rate on paid (basic search)	Credits / mo @ 50% paid mix	Fits Project ~4k?
5%	~3,750	Yes
10%	~7,500	No — Bootstrap ~\$100 or higher
25%	~18,750	No — Startup/Growth-class

Keep mining off the RAG Project key. Prefer basic search (1 credit) and Wikipedia-first.

8. BigQuery

Phase	Cap / job	Realistic \$/req (when BQ runs)
Bronze	250 MiB	~\$0.002
Marts	100 MiB	~\$0.0007
Free	N/A	\$0

Code must enforce plan gates (Free: no BQ) and maximumBytesBilled; LIMIT does not cap bytes scanned.

9. Sensitivities

Change	Effect
R1 on Agri/Integrated only (locked)	Avoids full-platform ~\$2.83k/mo Scenario B
Integrated exceeds 2k soft cap	COGS can exceed \$49.99 ARPU
Paid Tavily hit rate 10%+	Exceed Project 4k credits
Share Project key with mining	RAG fallback fails mid-month
No BQ byte caps	BQ can spike toward ~\$0.03/req
Marts instead of bronze	Paid non-R1 and R1 paths both cheaper

10. Recommendations

- Price and sell against allowance-based unit economics (Section 5), not 5 req/user/day.
- Budget platform ~\$1.28–1.6k/mo depending on Agri/Integrated share; do not budget full R1.
- Product-enforce Free gates (no BQ, no Tavily), query allowances, and Integrated soft cap.
- Keep Tavily Project ~\$30/mo for RAG; monitor credits; mining on a separate key.
- After marts migration, refresh COGS downward (~\$1.09k/mo capacity floor before R1 mix).

11. Launch gaps (non-COGS)

- Auth / plan spoofing prevention
- Rate limits and spend caps per plan
- Free BQ/Tavily gates in code
- Streamlit private (dev/QA only)
- Ingestion, backup, legal/retention

12. Disclaimer

Planning estimates. OpenRouter, Tavily, Railway, Qdrant, and BigQuery list prices change. Confirm Tavily Project packs on <https://www.tavily.com/pricing>. Validate BQ bytes, Railway memory, and Qdrant sizing before board lock.